

1. Given that

$$z_1 = 3 \left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right)$$

$$z_2 = \sqrt{2} \left(\cos\left(\frac{\pi}{12}\right) - i \sin\left(\frac{\pi}{12}\right) \right)$$

(a) write down the exact value of

(i) $|z_1 z_2|$

(ii) $\arg(z_1 z_2)$

(2)

Given that $w = z_1 z_2$ and that $\arg(w^n) = 0$, where $n \in \mathbb{Z}^+$

(b) determine

(i) the smallest positive value of n

(ii) the corresponding value of $|w^n|$

(3)

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Answer ALL questions. Write your answers in the spaces provided.

1.

$$f(z) = z^4 + az^3 + bz^2 + cz + d$$

where a , b , c and d are real constants.

Given that $-1 + 2i$ and $3 - i$ are two roots of the equation $f(z) = 0$

(a) show all the roots of $f(z) = 0$ on a single Argand diagram, (4)

(b) find the values of a , b , c and d . (5)

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1.

$$f(z) = 3z^3 + pz^2 + 57z + q$$

where p and q are real constants.

Given that $3 - 2\sqrt{2}i$ is a root of the equation $f(z) = 0$

(a) show all the roots of $f(z) = 0$ on a single Argand diagram, (7)

(b) find the value of p and the value of q . (3)

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1.
$$f(z) = z^4 - 6z^3 + az^2 + bz + 145$$

where a and b are real constants.

Given that $2 + 5i$ is a root of the equation $f(z) = 0$

(a) determine the other roots of the equation $f(z) = 0$ (7)

(b) Show all the roots of $f(z) = 0$ on a single Argand diagram. (2)



7. Given that $z = a + bi$ is a complex number where a and b are real constants,

(a) show that zz^* is a real number.

(2)

Given that

- $zz^* = 18$

- $\frac{z}{z^*} = \frac{7}{9} + \frac{4\sqrt{2}}{9}i$

(b) determine the possible complex numbers z

(5)

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- (3)

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6. In an Argand diagram, the points A , B and C are the vertices of an equilateral triangle with its centre at the origin. The point A represents the complex number $6 + 2i$.
- (a) Find the complex numbers represented by the points B and C , giving your answers in the form $x + iy$, where x and y are real and exact.

(6)

The points D , E and F are the midpoints of the sides of triangle ABC .

- (b) Find the exact area of triangle DEF .

(3)

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4. (i) Given that

$$z_1 = 6e^{\frac{\pi}{3}i} \text{ and } z_2 = 6\sqrt{3}e^{\frac{5\pi}{6}i}$$

show that

$$z_1 + z_2 = 12e^{\frac{2\pi}{3}i} \quad (3)$$

(ii) Given that

$$\arg(z - 5) = \frac{2\pi}{3}$$

determine the least value of $|z|$ as z varies.

(3)

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5. The locus C is given by

$$|z - 4| = 4$$

The locus D is given by

$$\arg z = \frac{\pi}{3}$$

- (a) Sketch, on the same Argand diagram, the locus C and the locus D (4)

The set of points A is defined by

$$A = \{z \in \mathbb{C} : |z - 4| \leq 4\} \cap \left\{z \in \mathbb{C} : 0 \leq \arg z \leq \frac{\pi}{3}\right\}$$

- (b) Show, by shading on your Argand diagram, the set of points A (1)
- (c) Find the area of the region defined by A , giving your answer in the form $p\pi + q\sqrt{3}$ where p and q are constants to be determined. (4)



4. (a) Use de Moivre's theorem to prove that

$$\sin 7\theta = 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta \quad (5)$$

- (b) Hence find the distinct roots of the equation

$$1 + 7x - 56x^3 + 112x^5 - 64x^7 = 0 \quad (5)$$

giving your answer to 3 decimal places where appropriate.

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- (a) Show that

where n is a positive integer.

(2)

- (b) Show that

(5)

- (c) Hence, making your reasoning clear, determine all the solutions of

in the interval $0 \leq \theta < 2\pi$

(3)

4. The infinite series C and S are defined by

$$C = \cos \theta + \frac{1}{2} \cos 5\theta + \frac{1}{4} \cos 9\theta + \frac{1}{8} \cos 13\theta + \dots$$

$$S = \sin \theta + \frac{1}{2} \sin 5\theta + \frac{1}{4} \sin 9\theta + \frac{1}{8} \sin 13\theta + \dots$$

Given that the series C and S are both convergent,

(a) show that

$$C + iS = \frac{2e^{i\theta}}{2 - e^{4i\theta}} \quad (4)$$

(b) Hence show that

$$S = \frac{4\sin \theta + 2\sin 3\theta}{5 - 4\cos 4\theta} \quad (4)$$

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9. (a) Given that $|z| < 1$, write down the sum of the infinite series

$$1 + z + z^2 + z^3 + \dots \quad (1)$$

- (b) Given that $z = \frac{1}{2}(\cos \theta + i \sin \theta)$,

- (i) use the answer to part (a), and de Moivre's theorem or otherwise, to prove that

$$\frac{1}{2} \sin \theta + \frac{1}{4} \sin 2\theta + \frac{1}{8} \sin 3\theta + \dots = \frac{2 \sin \theta}{5 - 4 \cos \theta} \quad (5)$$

- (ii) show that the sum of the infinite series $1 + z + z^2 + z^3 + \dots$ cannot be purely imaginary, giving a reason for your answer.

(2)

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8. (a) Given

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad n \in \mathbb{N}$$

show that

$$32 \cos^6 \theta \equiv \cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10 \quad (5)$$

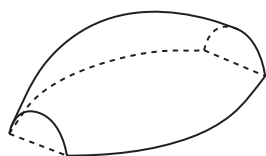


Figure 1

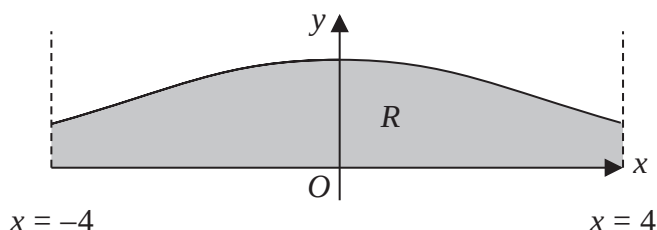


Figure 2

Figure 1 shows a solid paperweight with a flat base.

Figure 2 shows the curve with equation

$$y = H \cos^3 \left(\frac{x}{4} \right) \quad -4 \leq x \leq 4$$

where H is a positive constant and x is in radians.

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = -4$, the line with equation $x = 4$ and the x -axis.

The paperweight is modelled by the solid of revolution formed when R is rotated 180° about the x -axis.

Given that the maximum height of the paperweight is 2 cm,

(b) write down the value of H . (1)

(c) Using algebraic integration and the result in part (a), determine, in cm^3 , the volume of the paperweight, according to the model. Give your answer to 2 decimal places.

[Solutions based entirely on calculator technology are not acceptable.] (5)

(d) State a limitation of the model. (1)



